香港中文大學
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ROSE5710 Fundamental Robotics and Systems

Lecture Set 3: Robot Modelling Foundations

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2026-27

Lecture Summary

Aims of the lecture

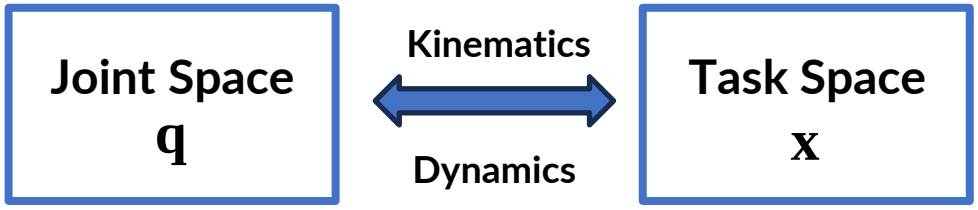
- To define the joint and task-space coordinates of robot manipulators
- To construct transformation matrices for general transformation of rigid bodies
- To represent the position and orientation for rigid bodies

References

- Robotics, Vision and Control, 3rd edition, Peter Corke, Springer, 2023

Joint and Task Space

Two important “spaces”





- What is joint space and how to define it?
- What is task space and how to define it?

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Rigid Robots are Comprised of Joints

Begin on understanding the robot joints

Revolute joint	1 DoF	$\mathbf{q} = [\theta]$
Prismatic joint	1 DoF	$\mathbf{q} = [x]$
Cylindrical joint	2 DoFs	$\mathbf{q} = [z \quad \theta]^T$
Universal joint	2 DoFs	$\mathbf{q} = [\theta_1 \quad \theta_2]^T$
Spherical joint	3 DoFs	$\mathbf{q} = [\theta_1 \quad \theta_2 \quad \theta_3]^T$
Spatial robot	6 DoFs	$\mathbf{q} = [x \ y \ z \ \theta_1 \ \theta_2 \ \theta_3]^T$





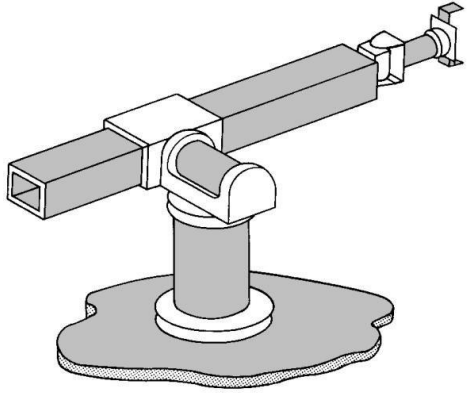


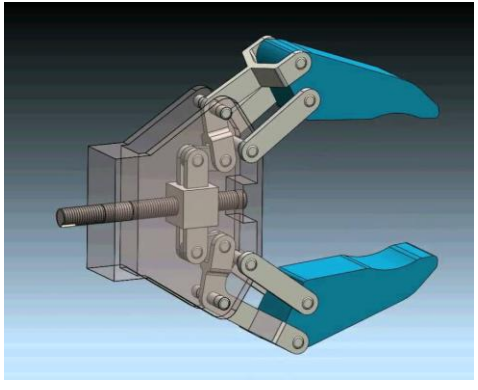


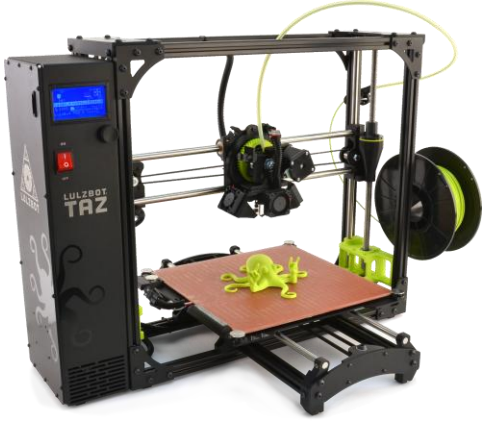


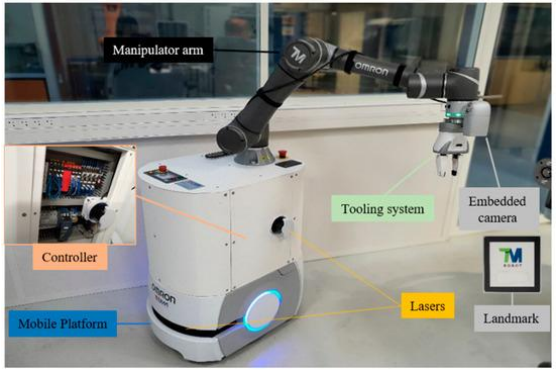
Defining the Joint Space









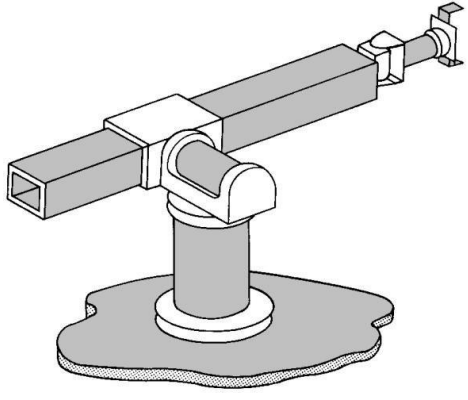


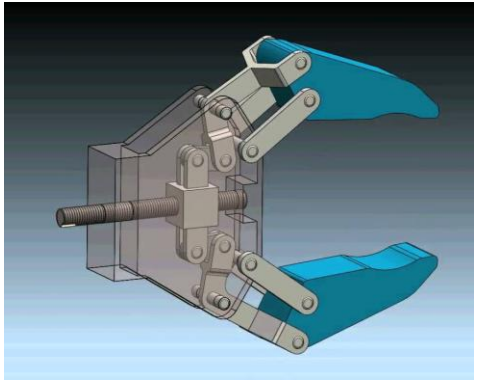


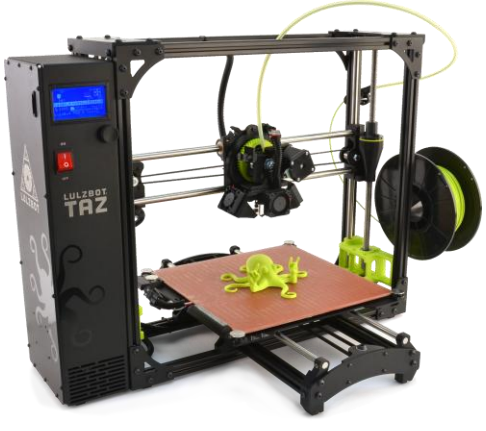
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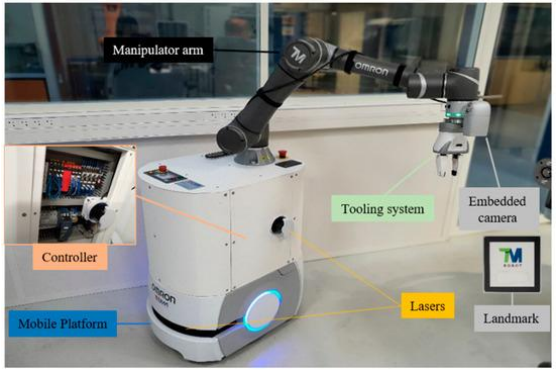
Defining the Task Space









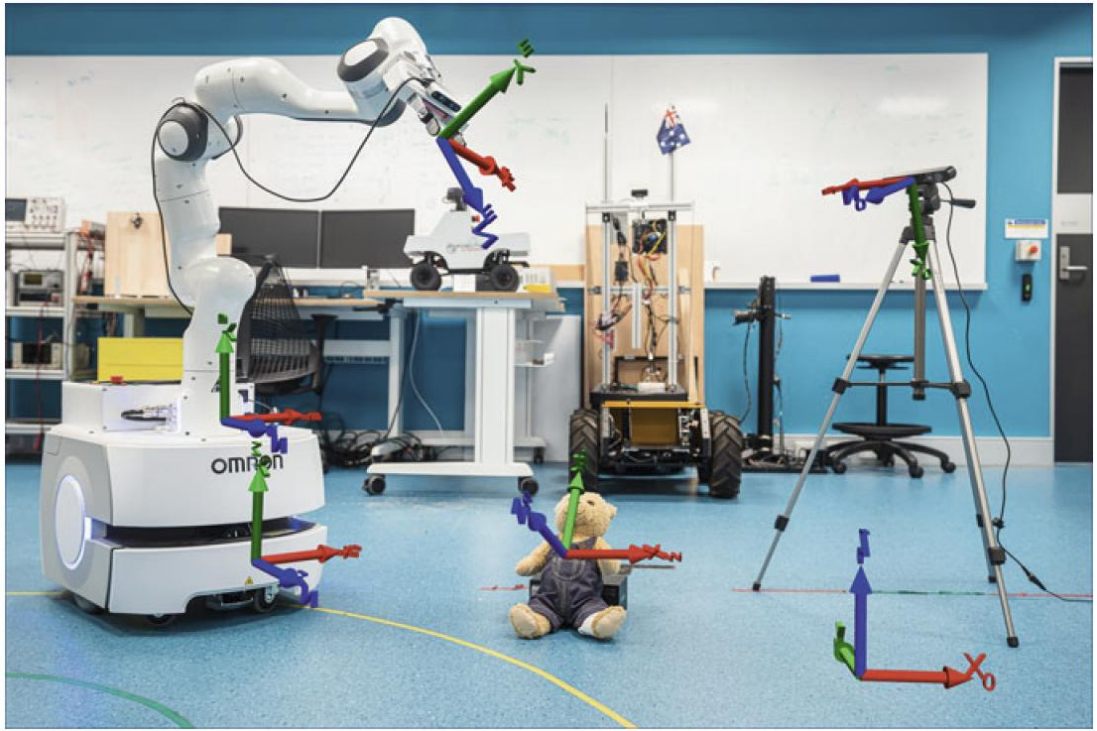




Why Do We Need Coordinate Systems?

The motion of points (or even rigid bodies) need vectors

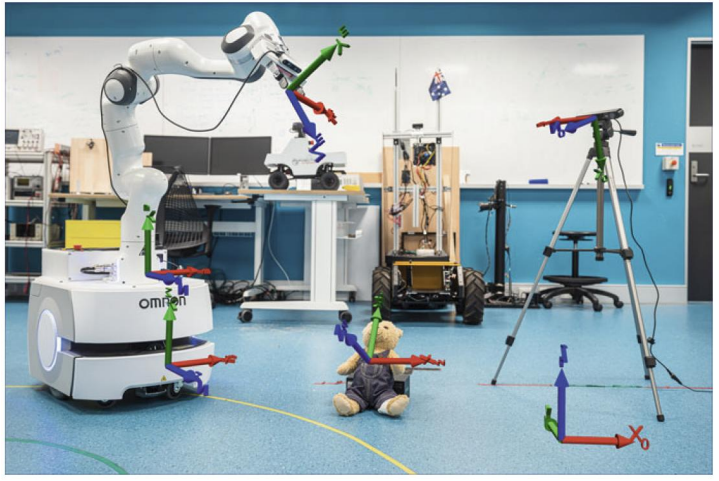
- Vectors need to be attached to a basis



Inertial and Non-Inertial Frames

Every point or vector must be represented in a coordinate system

$$^{a}\mathbf{r}_P = X\mathbf{i}_a + Y\mathbf{j}_a + Z\mathbf{k}_a$$
$$^{b}\mathbf{r}_P = A\mathbf{i}_b + B\mathbf{j}_b + C\mathbf{k}_b$$

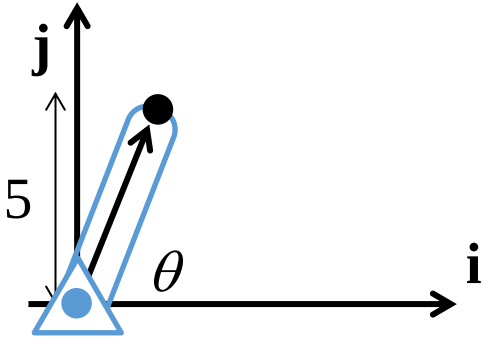


- Denote frames by $\{\mathcal{F}_a\},\{\mathcal{F}_b\}$

- **Inertial:** A coordinate system that is observed to be at rest, either does not move or zero acceleration (net force acting on it is zero)
- **Non-inertial:** coordinate system moves with non-zero acceleration

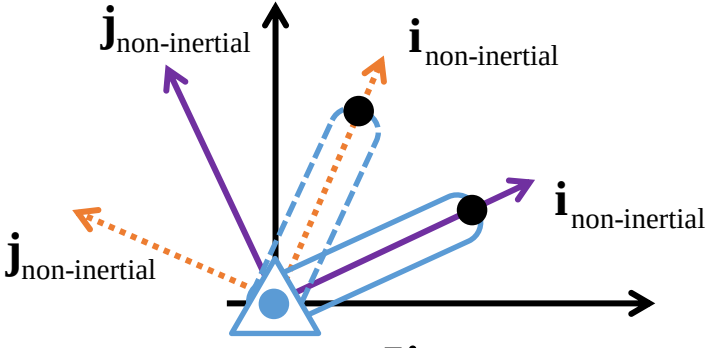
Why Non-Inertial Frames

Simple example



$$\mathbf{r}_P = 5\cos(\theta)\mathbf{i} + 5\sin(\theta)\mathbf{j}$$

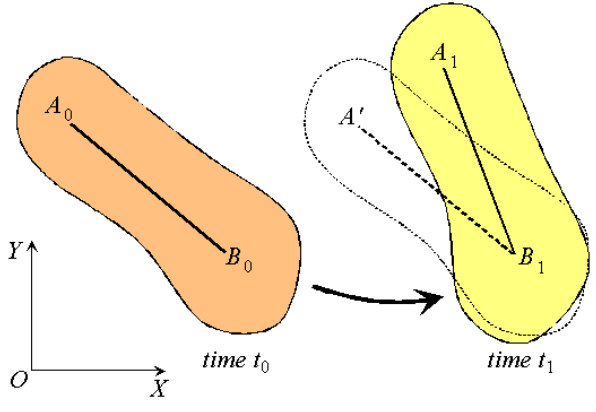
Inertial frame



$$\mathbf{r}_P = 5\mathbf{i}_{\text{non-inertial}}$$

Non-inertial frame

- Useful for rigid body motion (robotics)

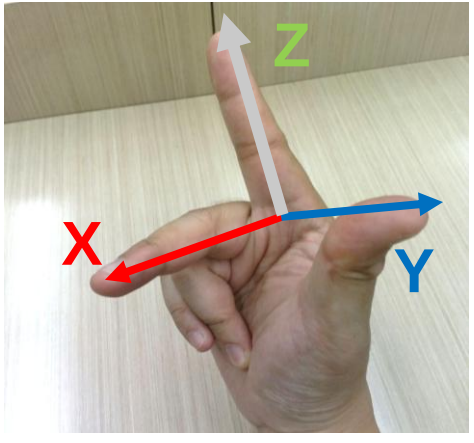


- Useful for multi-rigid-body motion

Defining Coordinate Systems in 3D

There are requirements on what a coordinate frame are

- For an *n* dimensional space $\mathbb{R}^n$ (*n* being 2 or 3), a set of *n* linearly **independent** vectors
- Orthonormal coordinate frame requires a set of linearly independent **orthonormal** vectors
- Right handed coordinate system



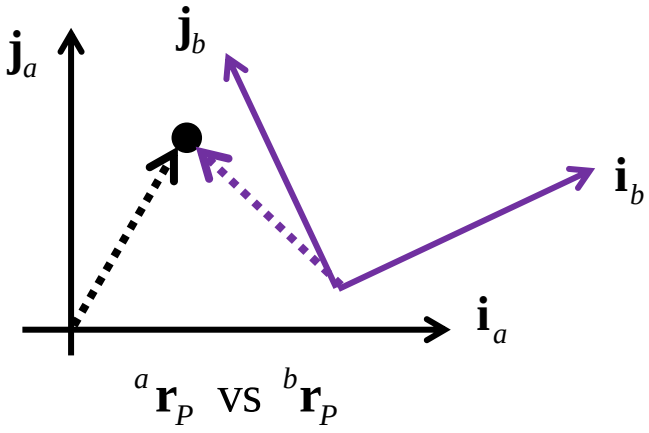
Mathematically

$$\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$$

- Orthogonal $\mathbf{x}_i \cdot \mathbf{x}_j = 0 \quad i \neq j$
- Normal vectors $\|\mathbf{x}_i\|^2 = 1 \quad \forall i$

Vectors in Non-Inertial Frames

Vectors in different frames



Important

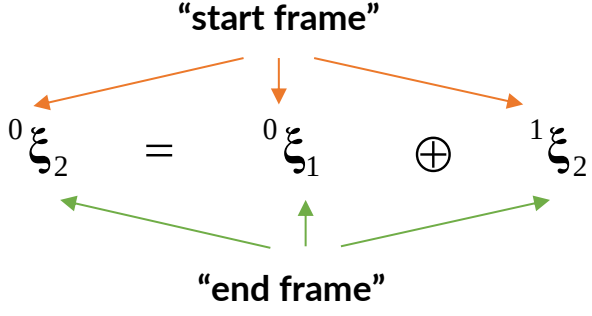
- ${}^a\mathbf{r}_P$ refers to the vector from origin to point *P* described by $\mathbf{i}_a, \mathbf{j}_a$
- ${}^a\mathbf{r}_P$ vs ${}^b\mathbf{r}_P$ refers to the same point E but described in $\mathbf{i}_a, \mathbf{j}_a$ vs $\mathbf{i}_b, \mathbf{j}_b$

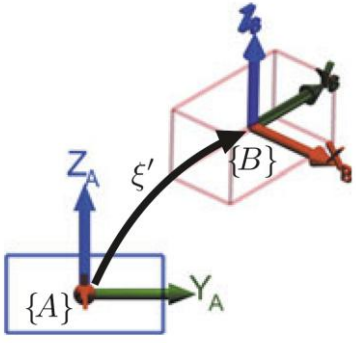
- **Different vector** but the **same point physically**

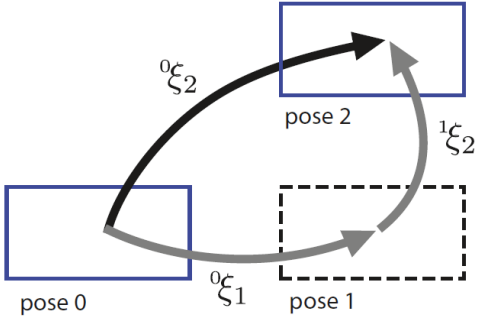
Frame Transformations

Denote the “motion” between poses/frames as ξ

- And when there’s multiple poses/frames







- Composition/compounding applies

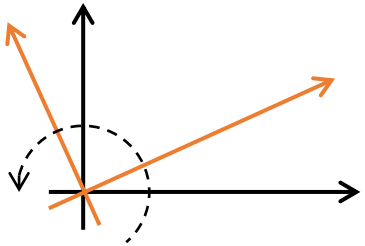
Other properties

- Null or no motion $^X\xi_Y \oplus {}^Y\xi_X = \varnothing$
- Inverse motion $^Y\xi_X \equiv \ominus\, {}^X\xi_Y$

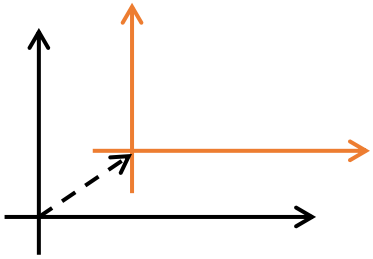
Types of Transformations

Three types of transformations

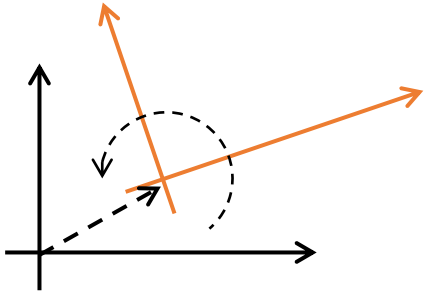
- Rotation
- Translation
- Translation + Rotation (general)



Rotation only



Translation only



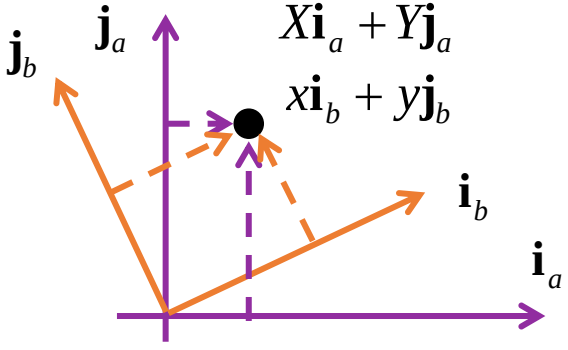
Translation + Rotation

Start in 2-D and then extend to 3-D

- Therefore 3 translation and 3 rotation coordinates (at least)

Rotation Matrix in 2D

The relationship between vectors in two frames ${}^a\mathbf{r}_P = \begin{bmatrix} X \\ Y \end{bmatrix}, {}^b\mathbf{r}_P = \begin{bmatrix} x \\ y \end{bmatrix}$



$$X = x\cos\theta - y\sin\theta$$
$$Y = x\sin\theta + y\cos\theta$$

$$\begin{bmatrix} X \\ Y \end{bmatrix} = \underbrace{\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}}_{R} \begin{bmatrix} x \\ y \end{bmatrix}$$

Define the rotation matrix ${}^a\mathbf{r}_P = {}^aR_b\,{}^b\mathbf{r}_P$

Special orthogonal group $R \in SO(2)$

Rotation Matrix in 2D

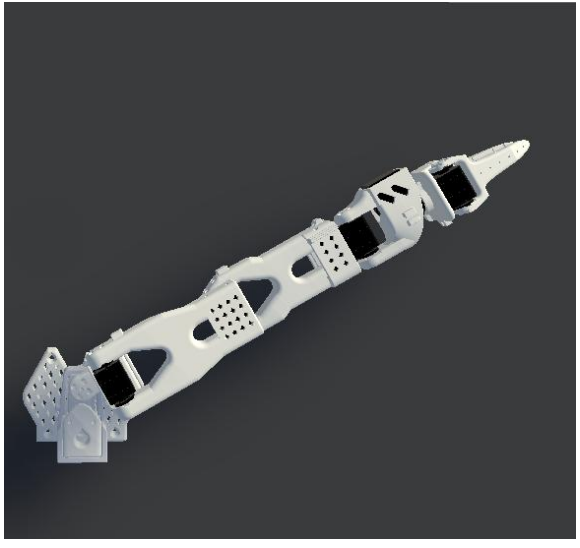
Example on robot



0 degrees rotation



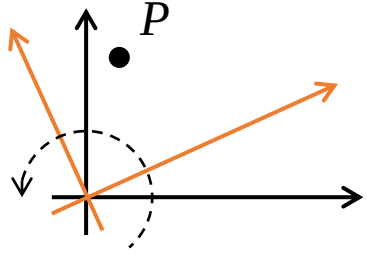
30 degrees rotation



-60 degrees rotation

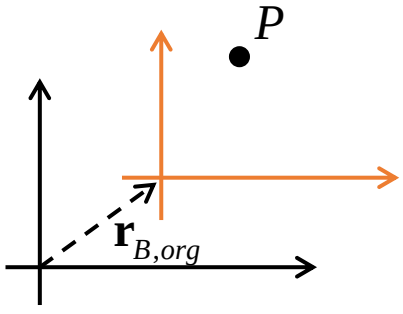
Rotations and Translations

Summarising



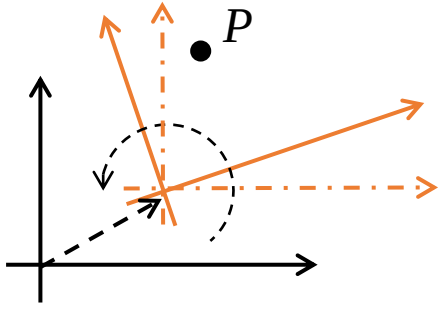
Rotation only

$${}^{a}\mathbf{r}_{P} = {}^{a}_{b}R\,{}^{b}\mathbf{r}_{P}$$



Translation only

$${}^{a}\mathbf{r}_{P} = {}^{a}\mathbf{r}_{B,org} + {}^{b}\mathbf{r}_{P}$$



Translation + Rotation

$${}^{a}\mathbf{r}_{P} = {}^{a}\mathbf{r}_{B,org} + {}^{a}_{b}R\,{}^{b}\mathbf{r}_{P}$$

3-D Transformations

From 2-D to 3-D

- The vector relationship remains the same, except all vectors and matrices are 3x1 or 3x3, respectively

$$ {}^{a}\mathbf{r}_{P} = {}^{a}\mathbf{r}_{BORG} + {}^{a}_{b}R\,{}^{b}\mathbf{r}_{P} $$

Translation part is straightforward, what about rotation?

- There are many different representations for 3-D rotation
- The most basic is the rotation matrix

$$ {}^{a}_{b}R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \in SO(3) $$

- A 9 value orthogonal matrix with determinant of 1

Special Orthogonal Group: *SO(n)*

Special subset of orthogonal group

- Distance-preserving transformations of a Euclidean space of dimension *n* that preserve a fixed point

$$AA^T = I_3, A \in SO(n)$$

For *SO(n)*:

- Being a group under the operation of matrix multiplication means that the product of any two matrices belongs to the group

$$A, B \in SO(n) \Leftrightarrow AB \in SO(n)$$

- As does its inverse

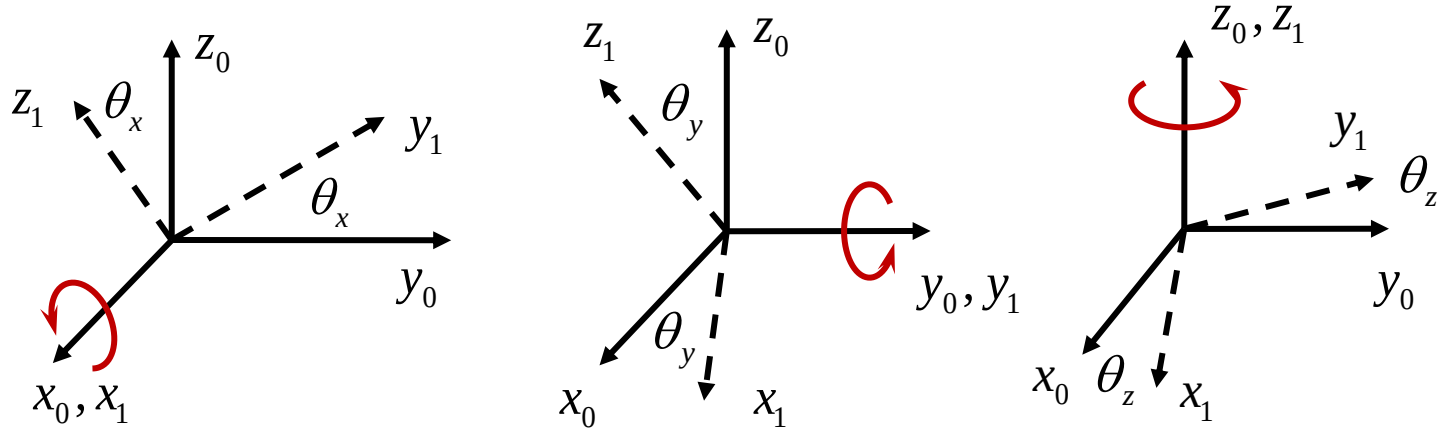
$$A \in SO(n) \Leftrightarrow A^T = A^{-1} \in SO(n)$$

- Associativity

$$(R_1 R_2) R_3 = R_1 (R_2 R_3),\ R_1, R_2, R_3 \in SO(n)$$

3-D Rotation Matrices

State with rotation about a single principle axis is a fundamental rotation

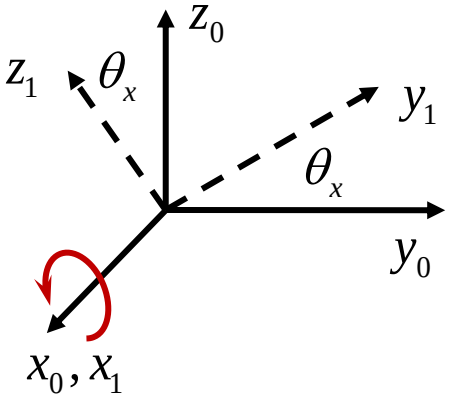


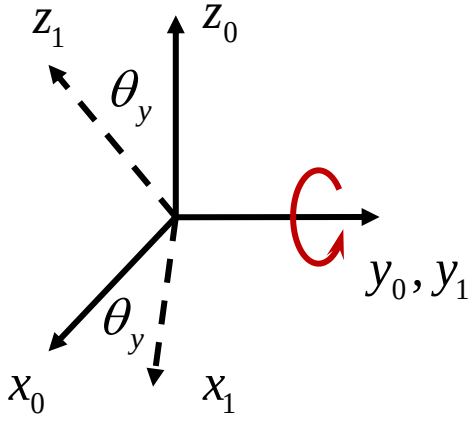
Note: Positive rotation is the "right hand rotation" about an axis!

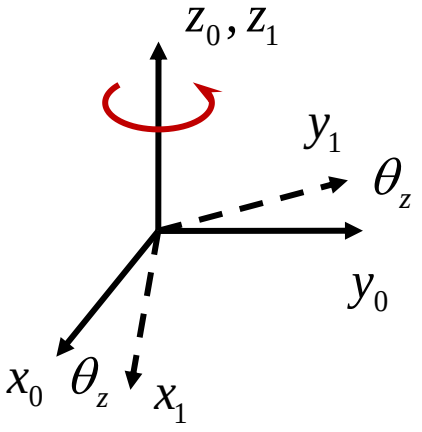
- Otherwise it is negative rotation

3-D Rotation Matrices

State with rotation about a single principle axis is a fundamental rotation







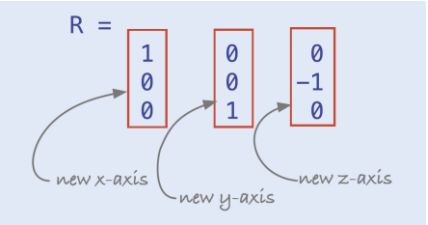
$$ {}^{0}_{1}R = \begin{matrix}\text{before frame}\\ \text{after frame}\end{matrix} R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_x & -\sin\theta_x \\ 0 & \sin\theta_x & \cos\theta_x \end{bmatrix} $$

$$ \begin{matrix}\text{before frame}\\ \text{after frame}\end{matrix} R = \begin{bmatrix} \cos\theta_y & 0 & \sin\theta_y \\ 0 & 1 & 0 \\ -\sin\theta_y & 0 & \cos\theta_y \end{bmatrix} $$

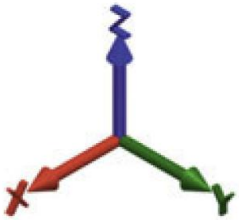
$$ \begin{matrix}\text{before frame}\\ \text{after frame}\end{matrix} R = \begin{bmatrix} \cos\theta_z & -\sin\theta_z & 0 \\ \sin\theta_z & \cos\theta_z & 0 \\ 0 & 0 & 1 \end{bmatrix} $$

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3-D Rotation Matrices



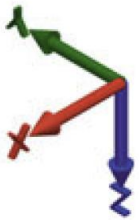
Examples



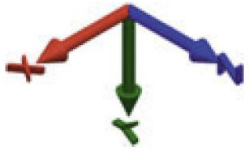
a Original



b $\frac{\pi}{2}$ about x-axis



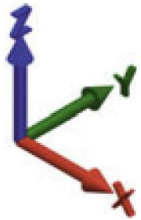
c π about x-axis



d $-\frac{\pi}{2}$ about x-axis



e $\frac{\pi}{2}$ about y-axis



f $\frac{\pi}{2}$ about z-axis

Inverse and Transpose of Rotation Matrix

The rotation matrix represents

$$ {}^{a}\mathbf{r} = {}^{a}_{b}R \, {}^{b}\mathbf{r} $$

- Pre-multiply the rotation matrix of a vector in frame {b} to same vector in frame {a}

So in reverse

- To convert a vector in frame {a} to {b}

$$ {}^{b}\mathbf{r} = {}^{b}_{a}R \, {}^{a}\mathbf{r} $$

$$ {}^{1}_{2}R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.866 & -0.5 \\ 0 & 0.5 & 0.866 \end{bmatrix} $$

- Where

$$ {}^{b}_{a}R = ({}^{a}_{b}R)^{-1} = ({}^{a}_{b}R)^{T} $$

$$ {}^{2}_{1}R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.866 & 0.5 \\ 0 & -0.5 & 0.866 \end{bmatrix} $$

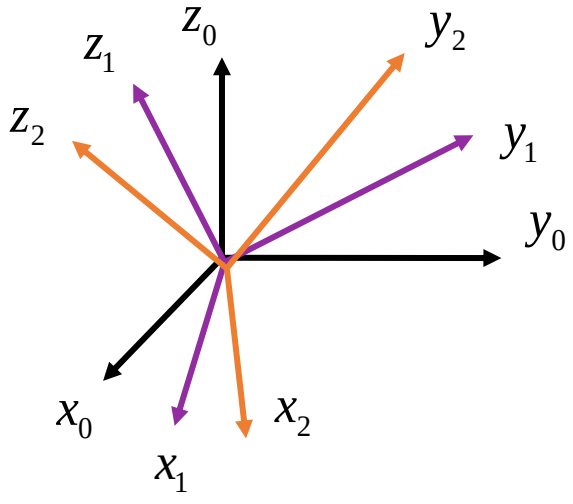
- This is since the rotation matrix is an orthogonal matrix with determinant of 1

Multiple Frame Rotations (Compound)

How to consider multiple rotations

$${}^{0}_{1}R, {}^{1}_{2}R$$

- How to find ${}^{0}_{2}R$?



Consider a rotation $\{2\} \rightarrow \{1\}$ then $\{1\} \rightarrow \{0\}$

$${}^{1}\mathbf{r} = {}^{1}_{2}R\,{}^{2}\mathbf{r} \quad \text{combined} \quad {}^{0}\mathbf{r} = {}^{0}_{1}R\,{}^{1}\mathbf{r} \longrightarrow {}^{0}\mathbf{r} = {}^{0}_{1}R\,{}^{1}_{2}R\,{}^{2}\mathbf{r}$$

- What does this become ${}^{?}\mathbf{r} = {}^{3}_{4}R\,{}^{4}_{5}R\,{}^{5}_{6}R\,{}^{6}_{7}R\,{}^{7}\mathbf{r}$

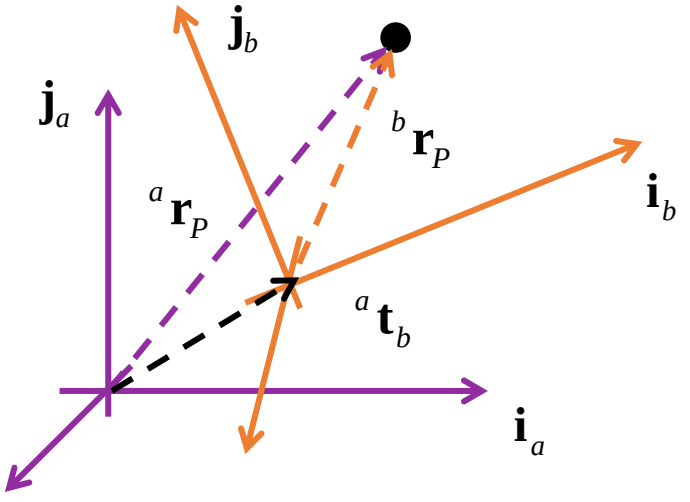
Homogeneous Transformations

For a translation and rotation transformation

- Also called a **general transformation**

$$ {}^{a}\mathbf{r}_P = {}^{a}\mathbf{t}_b + {}^{a}_{b}R\,{}^{b}\mathbf{r}_P $$

- Looks clumsy



Homogeneous transform is much cleaner

$$ {}^{a}\tilde{\mathbf{r}}_P = {}^{a}_{b}T\,{}^{b}\tilde{\mathbf{r}}_P $$

- Looks more elegant and hence appealing
- To achieve this, need to extend vectors to 4x1 and the transformation matrix T will be a 4x4 matrix

Homogeneous Transformations Matrix

Homogeneous transformation

$$ {}^{a}\mathbf{r}_{P} = {}^{a}\mathbf{t}_{b} + {}^{a}_{b}R\,{}^{b}\mathbf{r}_{P} $$

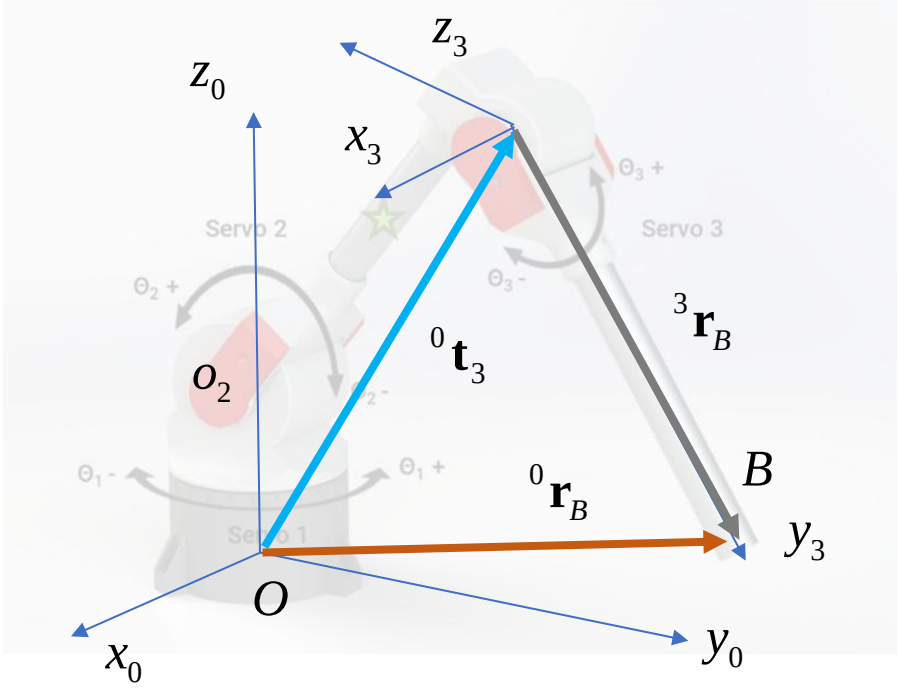
- Can be equivalently expressed as

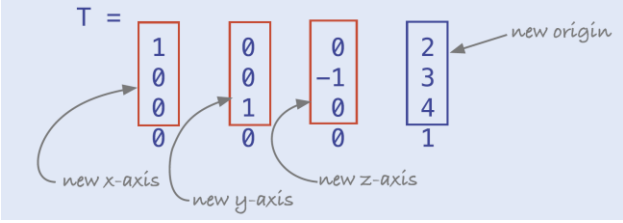
$$ \underbrace{\begin{bmatrix} {}^{a}\mathbf{r}_{P} \\ 1 \end{bmatrix}}_{{}^{a}\tilde{\mathbf{r}}_{P}} = \underbrace{\left[\begin{array}{ccc|c} & {}^{a}_{b}R & & {}^{a}\mathbf{t}_{b} \\ \hline 0 & 0 & 0 & 1 \end{array}\right]}_{{}^{a}_{b}T} \underbrace{\begin{bmatrix} {}^{b}\mathbf{r}_{P} \\ 1 \end{bmatrix}}_{{}^{b}\tilde{\mathbf{r}}_{P}} $$

- This encapsulates both the translation and rotation within a **single 4x4 transformation matrix**
- Usually no need to write the 3x1 vs 4x1 position vector differently, and only known by the implication of the T or R matrix

Homogeneous Transformations Matrix

Example on robot





$$\begin{bmatrix} {}^0\mathbf{r}_B \\ 1 \end{bmatrix} = \overbrace{\begin{bmatrix} \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \\ \hline \times & \times & \times & \times \end{bmatrix}}^{{}^0_2T} \begin{bmatrix} {}^3\mathbf{r}_B \\ 1 \end{bmatrix}$$

$${}^0\mathbf{r}_B = {}^0\mathbf{t}_3 + {}^0_3R\,{}^3\mathbf{r}_B$$

Special Euclidian Group: *SE(3)*

Combines group of rotation and translation

$$SO(n) = \left\{ \begin{bmatrix} R & \mathbf{t} \\ 0 & 1 \end{bmatrix} : R \in SO(n), \mathbf{t} \in \mathbb{R}^n \right\}$$

Unlike rotation matrix

- Orthogonality does not hold $({}^a_b T)^T ({}^a_b T) \neq 1$ and $({}^a_b T)^{-1} \neq ({}^a_b T)^T$
- Inverse operator

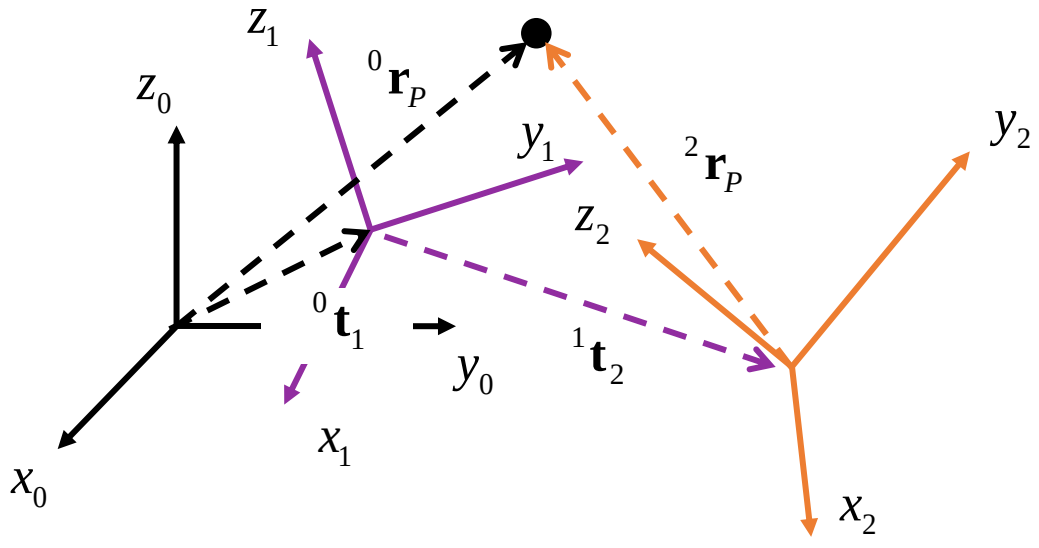
$${}^b_a T = ({}^a_b T)^{-1} = \left[\begin{array}{ccc|c} & {}^a_b R^T & & -{}^a_b R^T \, {}^a\mathbf{t}_b \\ \hline 0 & 0 & 0 & 1 \end{array} \right]$$

- Associativity

$$(T_1 T_2) T_3 = T_1 (T_2 T_3),\ T_1, T_2, T_3 \in SE(n)$$

Composition

Compound transform



- The resulting transform becomes

$$ {}^0_2T = {}^0_1T\,{}^1_2T = \begin{bmatrix} {}^0_1R\,{}^1_2R & {}^0\mathbf{t}_1 + {}^0_1R\,{}^1\mathbf{t}_2 \\ 0 \quad 0 \quad 0 & 1 \end{bmatrix} \equiv \begin{bmatrix} {}^0_2R & {}^0\mathbf{t}_2 \\ 0 \quad 0 \quad 0 & 1 \end{bmatrix} $$

Transformations in the Real World



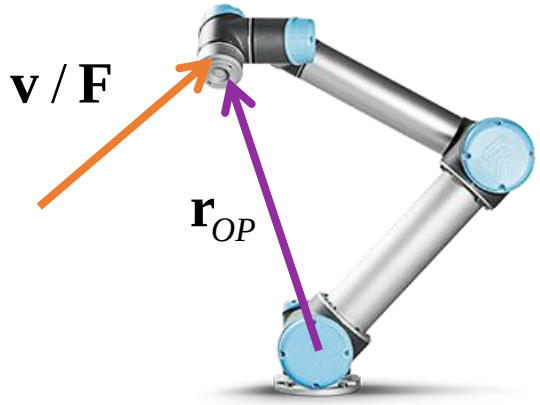
Important: Position Vector vs Free Vector

Position vector

- These vectors are defined from one point to another point

Free vector

- These vectors (such as velocity/force) only have magnitude and direction
- Cannot be used with transformation matrices!



$$ {}^{0}\mathbf{r}_P = {}^{0}_{1}T\,{}^{1}\mathbf{r}_P $$

$$ {}^{0}\mathbf{v}_P = {}^{0}_{1}R\,{}^{1}\mathbf{v}_P $$

~~$$ {}^{0}\mathbf{v}_P = {}^{0}_{1}T\,{}^{1}\mathbf{v}_P $$~~

Orientations in 3D

There are many representations that can be used for spatial/3-D orientation

- Euler/fixed angles (3 variables)
- Rotation matrix (9 variables)
- Axis-angle representation (4 variables)
- Quaternion (4 variables)

Advantages and disadvantages

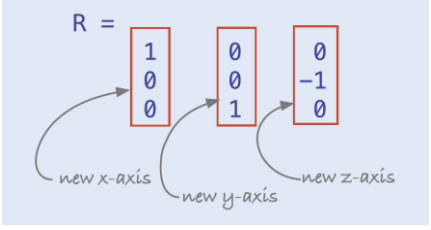
- Storage
- Singularities
- Ease of use (e.g. multiple rotations)

Rotation Matrix

The rotation matrix is a 3x3 matrix consisting of 9 values

- However, with 6 constraints since the matrix must be orthonormal

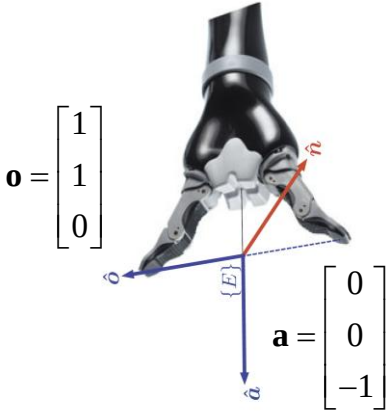
$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$



- Hard to "create" the rotation matrix

Two-vector representation

$$R = \begin{bmatrix} n_x & o_x & a_x \\ n_y & o_y & a_y \\ n_z & o_z & a_z \end{bmatrix} \qquad \hat{\mathbf{n}} = \hat{\mathbf{o}} \times \hat{\mathbf{a}}$$



$$R = \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

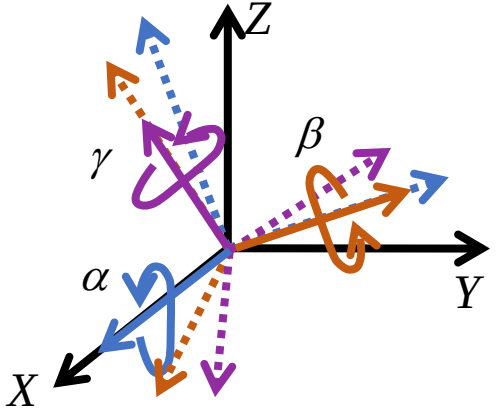
Three-Angle Representations

Three angles α, β, γ used to represent three successive rotations

- This representation uses the minimum number of variables to represent orientation (3 DoF with 3 variables)
- Example: *xyz*-Euler angle means rotation about x, then the new y then the new z

Advantages and disadvantages

- Definitions are not unique (12 different representations)
  - xyz, xyx, xzy, xzx, yxz, yzx, yzy, yxy, zxy, zyx, zyz, zxz
- Ambiguous: roll-pitch-yaw could mean ZYX or XYZ
- Gimbal lock $R_y(\beta)R_z(\pi/2)R_x(\gamma)$



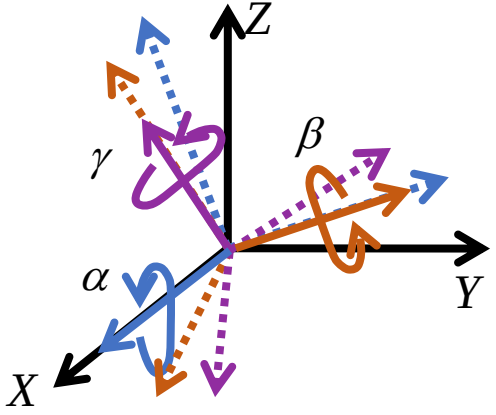
Three-Angle Representations: Euler-xyz Example

Assuming first rotation in x, then y, then z α, β, γ

$$R_z(\gamma) = \begin{bmatrix} \cos\gamma & -\sin\gamma & 0 \\ \sin\gamma & \cos\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad R_y(\beta) = \begin{bmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix} \quad R_z(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha \\ 0 & \sin\alpha & \cos\alpha \end{bmatrix}$$

Combine to obtain the final rotation

$${}^0_1R = R_z(\gamma)R_y(\beta)R_x(\alpha) = \begin{bmatrix} c\beta c\gamma & -c\beta s\gamma & s\beta \\ c\alpha s\gamma + s\alpha s\beta c\gamma & c\alpha c\gamma - s\alpha s\beta s\gamma & -s\alpha c\beta \\ s\alpha s\gamma - c\alpha s\beta c\gamma & s\alpha c\gamma + c\alpha s\beta s\gamma & c\alpha c\beta \end{bmatrix}$$

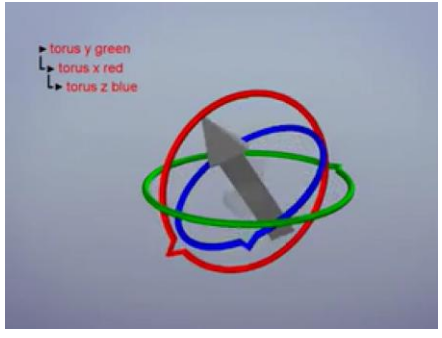


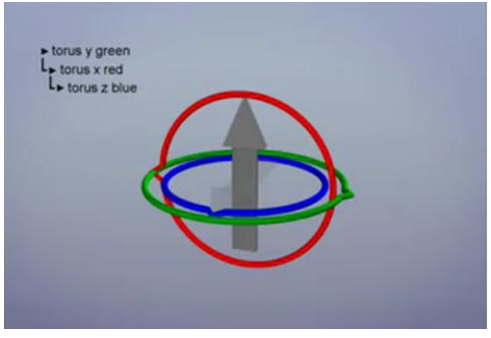
Three-Angle Representations: Singularities

In robotics, singularities is when one or more degrees of freedom is lost

- For example:

$$R_y(\beta)R_z(\pi/2)R_x(\gamma)$$



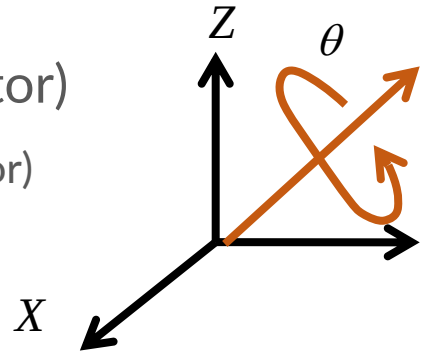


- The first and last rotation produce the same orientation

Axis-Angle Representation

Represent a rotation about an axis (unit vector)

- Requires 4 variable (one constraint for unit vector)
- Successive rotations are hard to represent



$$\hat{\mathbf{a}} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{z}$$

$$R = c\theta \cdot I_3 + s\theta\left[\hat{\mathbf{v}}\right]_\times + (1 - c\theta)(\hat{\mathbf{v}}\hat{\mathbf{v}}^T)$$

$$= \begin{bmatrix} a_x a_x v\theta + c\theta & a_x a_y v\theta - a_z s\theta & a_x a_z v\theta + a_y s\theta \\ a_x a_y v\theta + a_z s\theta & a_y a_y v\theta + c\theta & a_y a_z v\theta - a_x s\theta \\ a_x a_z v\theta - a_y s\theta & a_y a_z v\theta + a_x s\theta & a_z a_z v\theta + c\theta \end{bmatrix}$$

$$c\theta = \cos\theta, \quad s\theta = \sin\theta, \quad v\theta = 1 - \cos\theta$$

Quaternion

Hypercomplex number with

$$\breve{\mathbf{q}} = s + v_x i + v_y j + v_z k := (s, \mathbf{v}) \in \mathbb{H}$$

- Where

$$i^2 = j^2 = k^2 = ijk = -1$$

$$ij = -ji = k \qquad jk = -kj = i \qquad ki = -ik = j$$

The advantages of quaternions are

- Free of gimbal lock
- Convenient for generating smooth trajectories in orientation space
- Like rotational matrices, successive rotations can be better handled
- But less susceptible to round-off errors and more compact

Quaternion

Some quaternion operations

$$\breve{\mathbf{q}} = s + v_x i + v_y j + v_z k := (s, \mathbf{v})$$

- Conjugation

$$\breve{\mathbf{q}}^* = s - v_x i - v_y j - v_z k \in \mathbb{H}$$

- Multiplication

$$\breve{\mathbf{q}}_1 \circ \breve{\mathbf{q}}_2 = (s_1 s_2 - \mathbf{v}_1 \cdot \mathbf{v}_2 \quad , \quad s_1 \mathbf{v}_2 + s_2 \mathbf{v}_1 + \mathbf{v}_1 \times \mathbf{v}_2) \in \mathbb{H}$$

$$\breve{\mathbf{q}}_1 \circ \breve{\mathbf{q}}_2 \neq \breve{\mathbf{q}}_2 \circ \breve{\mathbf{q}}_1$$

- Dot product

$$\breve{\mathbf{q}}_1 \cdot \breve{\mathbf{q}}_2 = (s_1 s_2) + (v_{1x} v_{2x}) i + (v_{1y} v_{2y}) j + (v_{1z} v_{2z}) k$$

- Magnitude

$$\|\breve{\mathbf{q}}\| = \sqrt{\breve{\mathbf{q}} \cdot \breve{\mathbf{q}}} = \sqrt{s^2 + v_x^2 + v_y^2 + v_z^2}$$

- Unit quaternion (like axis-angle)

$$\tilde{\mathbf{q}} = \cos\frac{\theta}{2} + \left(v_x \sin\frac{\theta}{2}\right) i + \left(v_y \sin\frac{\theta}{2}\right) j + \left(v_z \sin\frac{\theta}{2}\right) k$$

Quaternion

Conversion to rotation matrix

$$R = \begin{bmatrix} 1-2(v_y^2+v_z^2) & 2(v_x v_y - s v_z) & 2(v_x v_z + s v_y) \\ 2(v_x v_y + s v_z) & 1-2(v_x^2+v_z^2) & 2(v_y v_z - s v_x) \\ 2(v_x v_z - s v_y) & 2(v_y v_z + s v_x) & 1-2(v_x^2+v_y^2) \end{bmatrix}$$

Conversion to axis-angle

$$\mathbf{a} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$$

$$\theta = 2\arctan(\|\mathbf{v}\|, s)$$

Rotation of vector $\mathbf{p}$

$$(0, {}^1\mathbf{p}) = \breve{\mathbf{q}} \circ (0, {}^0\mathbf{p}) \circ \tilde{\mathbf{q}}$$

Summary of Different Representations

Type	N	N_{min}	N_{min}/N
SO(2) matrix	4	1	25%
SE(2) matrix	9	3	33%
SE(2) as 2×3 matrix †	6	3	50%
2D twist vector	3	3	**100%**
SO(3) matrix	9	3	33%
Unit quaternion	4	3	75%
Unit quaternion vector part	3	3	**100%**
Euler parameters	3	3	**100%**
SE(3) matrix	16	6	38%
SE(3) as 3×4 matrix †	12	6	50%
3D twist vector	6	6	**100%**
Vector + unit quaternion	7	6	86%
Unit dual quaternion	8	6	75%